Rajesh Perumal Varadaraji

Location: Austin, TX **Email:** v.rajeshperumal@gmail.com **LinkedIn:** linkedin.com/in/vrajeshperumal

PROFESSIONAL SUMMARY

Senior software engineer with 18+ years building products end to end, from customer-facing React interfaces to cloud-native services and production agentic AI on AWS. At AWS Marketplace, built the MCP server that now powers the MAX purchasing agent, a citation-grounded EULA agent backed by a knowledge graph on Amazon Neptune, and led the unified Composable Procurement purchase flow shipped to every Marketplace customer. Focused on simplifying complex workflows for customers and shipping them safely with staged rollouts, observability and automated testing.

PROFESSIONAL EXPERIENCE

Amazon Web Services (AWS), Amazon Inc. – Austin, TX

Software Development Engineer II Jun 2022 - Present

- Proposed an AI-powered checkout and took it from prototype to production: built a Model Context Protocol (MCP) server on Amazon Bedrock AgentCore that lets AI assistants discover, configure and purchase AWS Marketplace offers, with cross-account authorization (FAS, KMS, STS AssumeRole) and deterministic, state-driven purchase triggers. It now powers the MAX agent in production.
- Architected and built the EULA Agent, which answers buyer questions about AWS Marketplace license agreements with citations to the exact clause. Combined OpenSearch hybrid retrieval (BM25 + k-NN, Titan embeddings) with a legal-ontology knowledge graph on Amazon Neptune (SPARQL), merged with Reciprocal Rank Fusion to surface exceptions, terminations and cross-references that vector search alone misses.
- Owned the EULA Agent’s 7-stage ingestion pipeline on Step Functions, EventBridge and Lambda: ontology-constrained LLM triple extraction, validation, embedding generation and bulk loading into Neptune.
- Led full-stack delivery of Composable Procurement, replacing two separate purchase flows with one unified React, TypeScript and XState session-based workflow, rolled out in stages to all Marketplace customers with zero downtime.
- Migrated procurement services and UI widgets to dualstack (IPv4/IPv6) across CDK packages, test frameworks and browser configurations in two weeks, ahead of the AWS Console IPv6 launch, with no impact on IPv4 customers; also delivered procurement routing work.
- Built an asynchronous post-purchase step in Go on Lambda (chosen for cold-start performance), triggered by the purchase event, with a UI status component and completion emails to buyers; wrote Go tooling for multi-region AWS CDK deployments.
- Raised engineering standards: enabled security scanning on CI/CD (static analysis at code review plus post-merge policy enforcement), defined the team’s ESLint error-versus-warning configuration and spec-driven rules for its AI coding assistant, and built CloudWatch structured logging, per-tool alarms and pytest coverage including cross-account trust-policy integration tests. Mentored junior engineers on React patterns and testing.

The Home Depot Inc. – Austin, TX

Senior Software Engineer Aug 2021 - Jun 2022

- Led a platform modernization initiative, establishing CI/CD pipeline standards and reusable HTTP client libraries that improved consistency, maintainability and service-integration patterns across engineering teams.
- Partnered with application teams to adopt standardized build, test, release and API-integration practices, and developed distributed services in Java and C# for retail and e-commerce operations.

Dell Technologies – Round Rock, TX

Principal Software Engineer Jun 2016 - Aug 2021

- Led the migration of monolithic applications to cloud-native microservices on Pivotal Cloud Foundry, integrating front-end applications, backend APIs and product-catalog services.
- Built full-stack e-commerce modules with ASP.NET Core, Angular, Bootstrap and REST APIs, including configurable-product pages, and re-engineered product-catalog definitions for configurable items.
- Integrated performance testing into the delivery pipeline to improve release confidence; worked with Kubernetes for containerized services.

Valence Health Inc. – Austin, TX

Senior .NET Developer Jun 2014 - May 2016

- Built claims-processing and member-enrollment applications using ASP.NET MVC 4 and REST APIs for healthcare operations workflows.

Tata Consultancy Services (client: Humana Inc.) – Louisville, KY and Chennai, India

Lead Software Engineer Nov 2008 - May 2014

- Built single-page applications with role-based security and WCF SOAP services supporting Humana's enterprise healthcare workflows (Louisville, 2012-2014).
- Developed applications across the full software development lifecycle (Chennai, 2008-2012).

EDUCATION

- Bachelor of Technology, Information Technology – Anna University, Chennai, India (2004-2008)

SKILLS

- **AI & agents:** Amazon Bedrock AgentCore, Model Context Protocol (MCP), Strands Agents SDK, LLM applications, RAG, tool calling, citation grounding, guardrails, human-in-the-loop workflows, Claude Code, Kiro
- **Search & knowledge graphs:** Amazon Neptune, SPARQL, RDF/OWL ontologies, OpenSearch hybrid retrieval (BM25 + k-NN), Titan embeddings, Reciprocal Rank Fusion; Neo4j/Cypher and Stardog (prototyping and ontology experiments)
- **Front end:** React, TypeScript, JavaScript, XState, Redux, Next.js, Angular, HTML5/CSS3, AWS Cloudscape, Jest, React Testing Library
- **Back end & languages:** Node.js, Python, Go, Java, C#, .NET Core, REST APIs, GraphQL, event-driven architecture, microservices
- **AWS & infrastructure:** Lambda, Step Functions, EventBridge, S3, DynamoDB, Fargate, IAM, KMS, STS, CloudWatch, AWS CDK, Docker, Kubernetes, Linux/Unix, dualstack IPv4/IPv6
- **Data:** SQL Server, PostgreSQL, DynamoDB, MongoDB
- **Delivery:** CI/CD, security scanning, staged rollouts, feature flags, pytest, Playwright, Cypress, Selenium, Git